PERFORMANCE WORK STATEMENT
BIOLOGICAL AND BENTHIC HABITAT SURVEYS
IN SUPPORT OF MAINTENANCE DREDGING

I. INTRODUCTION

A. General

The Contractor shall perform various biological and benthic studies to document benthic habitats, coral density, quality and cover, biofouling communities, invasive species, fish species, and protected species within the dredge footprint. The Contractor shall also provide recommendations for which corals found within the dredge footprint can potentially be successfully translocated. The period of performance for this work, including all deliverables, is 9 months.

B. Objective

The Contractor shall provide services for completing the work identified in this task order Performance Work Statement (PWS). An objective of the work to be performed is to enable compliance with the *Endangered Species Act* (ESA), *Marine Mammal Protection Act* (MMPA), and *Magnuson-Stevens Fishery Conservation and Management Act* (MSA). Additionally, these surveys will support compliance with *National Environmental Policy Act* (NEPA), Section 401 Water Quality Certification (WQC), and U.S. Army Corps of Engineers (USACE) permitting requirements under the *Clean Water Act* (CWA).

C. Background

Maintenance dredging is required to sustain operational depth within harbor waters and will have direct and indirect effects for the marine environment. The potential for that level of effect necessitates a high-resolution survey of benthic marine resources in the proposed dredge footprint.

For the purposes of this PWS, the Navy is primarily interested in marine resources that could be affected, either directly or indirectly, by the planned maintenance dredge. The PWS describes biological and benthic surveys and reports to ensure compliance with the ESA, MMPA, and MSA. Additionally, these surveys will support compliance with NEPA, Section 401 Water Quality Certification (WQC), and USACE permitting requirements under the CWA. This PWS also supports the Integrated Natural Resource Management Plan (INRMP) for the installation, as well as environmental planning for future construction projects and activities that could result in bottom disturbance in or near the project areas.

Federal government environmental mandates require all action proponents (i.e., the Navy) to use the best available science to conduct an assessment of the potential impacts on marine resources from performing in-water work. It is anticipated that during dredging, some corals, seagrasses, filtering communities, and/or other sensitive marine species will be directly and/or indirectly impacted in or near the footprint of the project. The surveys proposed in these task orders will characterize these marine resources and will support plans that the Navy may consider to avoid, minimize, and mitigate the impacts caused by the in-water, dredging activities.

Tasks in this PWS indicate the basic information needed and potential methods for data acquisition and presentation. The Contractor is encouraged to provide their recommended methods to accomplish the work described in this PWS effectively and efficiently.

~~**Previous Surveys**~~

~~Table 1 lists some relevant scientific surveys undertaken in the vicinity of the proposed maintenance dredging area and the areas adjacent to the projects presented in this PWS.~~

~~Table 1. Summary of recent benthic and biological surveys undertaken within or adjacent to the Project Area.~~

~~Survey Title~~	~~Location~~	~~Completion Date~~	~~Undertaken by~~

II. PROJECT POINTS OF CONTACT AND AGENCY COORDINATION

A. Task Order Contracting Officer's Representative (TOCOR)

The TOCOR for this task order is XXX, who can be reached by telephone at XXX or via email at XXX. The alternate TOCOR is XXX at XXX or via email at XXX. The Contractor shall keep the TOCOR and alternate TOCOR informed of progress and problems encountered. The TOCOR will serve as the Contractor point of contact on all technical matters associated with this contract

B. Contractor Project Manager

The Contractor shall designate one individual employee (by name and title) who shall be the central point of contact for matters pertaining to this contract. The Contractor shall notify the TOCOR and alternative TOCOR of the designated individual at time of submittal of the proposal.

C. Agency Coordination

The Navy will coordinate with local, state or Federal agencies as needed. The Contractor will assist as needed with pertinent agency coordination, including preparation of suggested memorandum or letterhead correspondence between the Navy and agencies. Deliverables produced under this task order shall list, by name, coordination and contact made with agencies.

D. Contract Administration.

The Contractor shall receive direction on all elements of this task order from Ms. Casey Sugihara and Mr. Kristopher Tom, Contract Administrators (CA). Correspondence should be addressed to:

Naval Facilities Engineering Command Pacific
Attn: XXX

258 Makalapa Drive, Suite 100
Pearl Harbor, HI 96860
Email: XXX

III. TASK SUMMARY

The Contractor shall perform various biological and benthic habitat surveys to document benthic habitats, coral density, quality, and cover, biofouling communities, invasive species, fish species, and protected species. The Contractor shall also provide recommendations for which corals found within the proposed dredge footprint can potentially be successfully translocated. These studies will inform the ESA and EFH consultations and aid in NEPA compliance and USACE permit approval. Special attention shall be paid to ensure that all data collected for this PWS are also useful to any future updates to the Navy's INRMP, as well as environmental planning for other future construction projects and activities that could result in bottom disturbance in or near the project areas.

The Contractor shall be responsible for the coordination of all work or services rendered and the professionalism and technical accuracy of all deliverables. Products submitted by the Contractor will represent the best data obtained or available and the most appropriate solutions to meet Federal

mandate requirements for environmental compliance. The Contractor is also encouraged to provide their recommended methods to accomplish the work described in this PWS effectively and efficiently. The Navy's TOCOR is the point of contact for discussions on the best approach for accomplishing work in this PWS.

A. Task 1: Kick-Off and Status Update Meetings

Within two weeks of task order award, the Contractor shall plan and lead a kick-off meeting between the Contractor and the Navy. It is anticipated that a one-hour, kick-off meeting between the Contractor and Navy's TOCORs will be necessary to discuss preliminary expectations and the requirements of the Navy's ESA and EFH Consultation Package for the proposed dredge. It is also anticipated that up to three, one-hour coordination conference calls will be needed to discuss any issues/questions that arise from either the Contractor or Navy. Additionally, monthly status update calls may be necessary throughout the duration of performance. The Contractor shall plan for at least this quantity and frequency of communication and shall include labor to cover this level of communication in the bid.

The Contractor shall provide meeting minutes from the kick-off meeting, as well as the up to three, one-hour coordination conference calls. Notes shall include the objective/purpose of the meeting, a list of participants, and the substantive outcome(s) of discussions. Minutes shall be submitted to the TOCOR within five working days following the meeting for review and approval. The Contractor shall resolve Navy comments and finalize the minutes within five working days of receiving Navy comments. The Contractor will finalize and distribute final minutes by e-mail to the TOCOR.

B. Task 2: Work Execution Plan, Dive Plan, and Health & Safety Plan

The Contractor shall prepare a Work Execution Plan for performing the biological surveys to document protected species, fish and benthic habitat distribution and cover. The Work Execution Plan shall include (but is not limited to) the following sections:

- Methods
- Data reporting and analysis
- Diving operations
- Deliverables
- Schedule of performance.

The Contractor shall also create both a Dive Plan and a Health and Safety Plan that are compliant with the *US Army Corps of Engineers Safety and Health Requirements Manual* (EM 385-1-1). The Plan shall be submitted to the TOCOR prior to field work for review. The plan will be reviewed and approved by installation Dive Master at least 10 working days prior to the contractors entering the water.

For all in-water activities (e.g., diving, anchoring, staging of materials, work with tools and transects, etc.), all divers will avoid physical contact with organisms, especially corals. Any materials being staged on the bottom should be thoroughly stabilized and secured to avoid movement, especially during time periods of elevated wave action, especially a typhoon risk. Divers should also avoid exposing corals directly or indirectly to toxico-pathological agents, especially through the use of reef safe sunscreen.

This survey requires the Contractor to visit military areas. Personnel assigned to the project must be able to qualify for access to military areas after appropriate security checks are made. All personnel shall abide

by prevailing security requirements during visits during visits to such military property and shall obtain the required permits and passes in a timely manner. When applicable, the Contractor is responsible for preparation and submittal of application for all necessary passes and entry permits for Contractor and sub-contractor's personnel and vehicles to enter Department of Defense installations.

C. **Task 3: Focused Biological Surveys**

The Contractor shall plan and execute appropriate methods of data collection agreed to during discussion with the TOCOR and NAVFAC Pacific subject matter experts (SME). Via a team of qualified scientific divers, the Contractor shall collect and analyze quantitative biological data using methodologies accepted by regulatory agencies to support environment compliance. Located in an active harbor in O'ahu, Hawai'I (Figure 1), the Contractor shall collect data across the 5 acre area: on the benthos adjacent to the wharf, and on the wharf face itself. Depths are expected to be no deeper than 60 feet.



Figure 1. Map of survey area.
note: map generated using Google Earth for example purposes only

The Contractor shall suggest methods (e.g., transects, photomosaics, drop camera photography, etc.) that are suitable for quantifying coral and sessile, filter-feeding invertebrates, and provide general estimates of fish and other mobile species (i.e., turtles, marine mammals, or mobile invertebrates). These quantitative surveys will occur in areas that will be directly and indirectly impacted by the proposed maintenance dredge. The main purpose of these surveys is to quantitatively determine the marine resources (including EFH) and protected species potentially affected by the proposed action, which in turn will be used to inform regulatory consultations pursuant to the ESA and MSA.

The Contractor shall record the following:

- A quantitative assessment of marine resources within the action footprint - the existing benthos, and on debris. A representative percentage of cover can be evaluated for extrapolation as recommended by field and biological experts.
- Species list of all corals observed to family and morphological level as possible
- Presence, location and quantity of ESA-listed species
- Size measurements of a representative subsample of coral colonies observed
- A quantitative evaluation on the health and condition of coral colonies observed
- Species list of fish observed
- Seagrass to the lowest taxonomic level possible
- Species list of invertebrates (sessile and mobile) observed besides coral
- List of macro algae observed
- A separate list of any invasive species that are observed
- Notes on any turtles or other large vertebrates observed
- Nature and quality of the natural substrate
- A description of anthropogenic debris and any marine species observed growing on debris (e.g., coral observed to be translocated, etc.)
- Georeferenced imagery (mosaics, photos, video, etc.) of the surveyed area.

Focused surveys shall be conducted to measure the spatial distribution of coral colonies, cover, species, colony sizes, density, growth form and general condition of hard and soft corals within the survey area. The Contractor shall strive to capture representative data of the direct and indirect impact areas. For purposes of this PWS, direct and indirect impact areas within the dredge footprint, as well as any underwater debris that may fall within the dredge footprint. Photographs of the resources shall be obtained if conditions are suitable in adjacent. As this area was previously dredged, and per results from previous surveys, a low number of marine resources are expected in most areas, but require verification. The Contractor shall survey all mooring devices and mooring chains, as well as any debris within the survey area.

Travel Requirements – Travel from out of the commuting area is not expected to be required for this project.

D. **Task 4: Deliverables**

4a. Report Writing - The Contractor shall compile and deliver the data and information assembled in Tasks 2 and 3. Following completion of all fieldwork under this task, a field summary presentation (call), a draft survey report, and a final survey report shall be prepared and document the results of the biological

surveys. The summary presentation shall briefly summarize any significant findings and imagery from the survey, and be provided within two weeks after completion of the survey fieldwork.

The draft and final survey reports will include, but not be limited to the following:

- Names of investigators and suggested citation
- Study purpose and location of project
- Materials and Methodology employed in the surveys
- A description of the current condition of the site(s), including location data (e.g., presence of organisms of interest, protected species sightings, coordinate location, water depth, etc.)
- Percent cover of benthic organisms/categories to be established during the planning meetings: hard coral, soft coral, seagrass, filtering community (bivalves, sponges, etc.), algae (macro, turf and crustose coralline algae), sand, silt, debris, and other notable marine resources.
- Quantitative measures of sessile fauna and flora, including hard and soft coral (species, abundance, size-frequency distributions, morphology), seagrass (extent and density), sand, algae, filter feeders (percent cover) etc.
- Presence, and where possible, quantitative estimates of abundance of fish and other mobile species
- Observations of indicators of growth, disease, potential biotic or abiotic threats/stressors to survivorship, invasive species, and overall health of corals.
- Relate approximate boundaries of features indicating major/general bottom substrate conditions, such as “unconsolidated sediment”, “artificial substrate”, “hard substrate/natural rock“ to resources found during the focused surveys
- Recommendations for coral translocations
- Project Area Map (including sampling locations)
- Provide geo-referenced habitat maps and coordinates of areas and resources of importance.

Task 4b. Orthographic Mosaics Raw and Processed Imagery - The Contractor shall provide all raw and processed imagery on external hard drives approximately eight weeks following completion of the final survey report. When processing and generating three-dimensional (3D) orthomosaic products for visual reference, the Contractor shall also provide the raw two-dimensional (2D) reconstructions (orthophotos) of the area surveyed that are used to georeference each orthoprojection. The Contractor shall also provide the orthoprojection, which includes a geometrically accurate 2D image of the reef from which subsequent analysis and data extraction can occur.

IV. ASSUMPTIONS INCLUDED IN THE STATEMENT OF WORK

E. Scope Exclusions

This PWS does not include authorizations for the Contractor to perform inherently governmental functions. Work may include performance of all efforts except those that may be considered inherently governmental.

F. State and Local Requirements

Since the proposed action to be analyzed under this PWS does not include areas under State and/or Local jurisdiction, compliance with any state or local requirements is only needed when required due to waiver of sovereign immunity.

G. **Government Furnished Information, Materials, Equipment, and Facilities**

1. The Government will furnish electronic copies of the documents referenced on page 1 of this PWS, if contractor does not have copies of the documents.
2. Aside from response to requests for information, comments on submittals or other information as described herein, the Government does not intend to furnish additional information. All information gathering is the responsibility of the Contractor.
3. The Government will not furnish any equipment or facilities for Contractor use during this effort. The Contractor shall provide the necessary hardware, software, personnel, materials, and facilities to perform the tasks specified by this PWS.

V. GENERAL STIPULATIONS

1. The Contractor responsibility for contract correspondence is directly to the Contracting Officer via the Contract Specialist. Any requested change/deviation in scope must be brought to the attention of and/or approved by the Contracting Officer. In no case will any person other than the Contracting Officer make changes to the contract scope. All changes to the work definition/period of performance shall be accompanied by modification to the contract by the Contracting Officer. Protocol for this contract task order is to always copy the TOCOR when corresponding with the contracting officer and contract specialist. With TOCOR concurrence on a case-by-case basis, the Contractor may correspond directly with other government personnel for the purposes of coordination, comment resolution, or other objectives related to performance of work within this task order. When corresponding with other government personnel, the Contractor shall always include the TOCOR on copy or as a required participant in verbal communication.

2. The Contractor is acting as an agent of the United States Navy and of the “Action Proponent” when performing work required of this contract. As an agent of the United States Navy and “Action Proponent” the Contractor is responsible for identifying and recommending any required conclusory statements required of reports of the United States Navy and “Action Proponent” in initial drafts of submittals for government review and requested concurrence.

3. **Performance Management.** The Contractor is responsible for proactive management of performance of work required by this task order.

4. **Professionalism/Conduct/Courtesy.** The Contractor is responsible to conduct its work in a professional and courteous manner, including appropriate attire, language, and protocol.

5. **Timeliness and Schedule Management.** The Contractor is responsible for proactive schedule management to include: establishment of schedule baselines; tracking of planned and actual dates; and schedule update/projection and communication of anticipated schedule changes.

6. **Technical Responsibility, Accuracy, and Sufficiency.**
   a. The Government will make available past reports/documents that may be relevant to the work to be performed in this task order, but it is the Contractor responsibility to conduct research

and obtain copies of potentially relevant documents for use/reference in documents prepared under this task order.

b. The Contractor is responsible for the technical accuracy and technical sufficiency of reports produced under this task order.
c. The Contractor must identify the source of information and date of information being used when information from a past or separate effort is utilized. The Contractor is responsible for providing copies of reference used and for preparation and verification of meeting notes and records of personal communication when used as sources of information.

7. **Quality Management.**
   a. The Contractor Quality Control Efforts shall ensure that submittals are complete documents, which have been reviewed for academic excellence, professional and copy quality, and technical accuracy. Submittals not displaying such efforts will not be accepted.
   b. Document Quality. The Contractor will arrange for printing, assembling, proofreading, editing, overall Quality Management and Quality Control, covering, binding, and distribution of each iteration of the submitted documents, as required.
   c. Three phases of quality management. The Contractor quality management/quality control shall incorporate building in quality control when planning work, during work performance, and quality control inspection by the Contractor prior to submittals sent for Government review.

8. **Neutral Point of View language for technical/scientific reports.** Submittals shall be written utilizing neutral point of view language, which means representing fairly, proportionately, and, as far as possible, without editorial bias, all of the significant views that have been published by reliable sources on a topic. Apply the following principles when using neutral point of view language:
   a. Avoid stating opinions as facts
   b. Avoid stating seriously contested assertions as facts
   c. Avoid stating facts as opinions
   d. Prefer non-judgmental language
   e. Indicate the relative prominence of opposing views.

9. **Writing Style standard and use of acronyms and abbreviations.** Contractor submittals shall follow writing style standard guidance and guidance for use of acronyms and abbreviations.

10. **Figures.** All figures in reports prepared under this task order must contain an accompanying reference in the text that summarizes the figure number being referenced, what the figure depicts, and relevance to the topic described in the text.

11. **Comment response matrix, line numbers, and red-line versions**.
    a. For all versions reviewed, following the initial version of a document, submit both “clean” as well as “red-lined” versions showing the revisions made to the previous version.
    b. All versions of a document following the initial version will be accompanied by a comprehensive comment response matrix that identifies each comment received previously and response action taken in preparation of the current submittal. All comments received on prior versions must be fully resolved prior to submittal of subsequent versions. All comments received on the pre-final must be fully resolved prior to submittal of the final.
    c. Fully resolved action/responses to comments should fall into one of three general categories: Concur, Exception or For Information Only (FIO).

i. Concur is the general category of response to comments providing factual correction that have been verified by the Contractor. Concur responses should refer to the specific location of existing, added, or revised text in the document.

ii. Exception is the general category of response to comments providing suggestion for revision or recommendation that have been verified by the Contractor to be either untrue or unadvised. Rationale should be provided for comments that the Contractor takes exception to.

iii. FIO is the general category of response to comments of inquiry. FIO responses should include reference to the specific location of text within the document that addresses the inquiry, or specific location and summary of text added /revised within the document that addresses the inquiry, or reason why no text needs be added or revised.

d. Responses to comments should be back-checked for acceptability with the individual providing the comments before being considered fully resolved in the comment/response matrix. Comments not yet fully resolved may be identified as pending, with proposed resolution general category and proposed response identified in the matrix. If no revisions within the document were made in response to any comment, the action/response indicated in the matrix should include a statement indicating so and providing rationale on why revisions were unnecessary. Questions and requests for information should be handled via separate correspondence prior to resolution or proposed resolution. General comment high-level back-brief for senior leadership of resolution of comments should normally be by exception only. Comment Resolution at a working-level should be able to use the matrix to focus on resolving pending responses and back-check responses for acceptability by the originator of the comment.

e. All Versions except the published version shall contain line numbers to facilitate review and commenting. The Contractor shall be responsible for the professional and technical accuracy and coordination of all data, reports, drawings and other work or materials furnished by the Contractor under this contract. All deliverables submitted to the Government by the Contractor are expected to have gone through a QA/QC process and review. All errors or deficiencies in assigned work attributable to the Contractor shall be corrected by the Contractor at no cost to the Government within one (1) week after discovery.

12. **Electronic Distribution of submittals**.
    a. All submittals shall at minimum be provided electronically.
    b. Electronic submittals provided for review shall be distributed by the Contractor directly to a distribution list via email. Email distribution list for any particular submittal shall be developed in coordination with the TOCOR no later than one week prior to submittal. For large file sizes, the Contractor may include a link to location where the submittals may be downloaded instead of attaching a file. When providing a link, the link must be from a system (such as AMRDEC Safe, or EIMS) that can be accessed via government computer systems. The Contractor shall include the following information in the text of the email when submitting documents for review electronically:
        i. Contract and Task Order Number
        ii. Title of submittal and reference to paragraph in this scope that requires the submittal.
        iii. Bulleted summary of the most important changes from previous version.

iv. Date by which comments are requested
v. To whom comments should be submitted
vi. Any requested format instructions for comments.

13. **This Task Order requires the Contractor to visit military areas**. Personnel assigned to the project must be able to qualify for access to military areas after appropriate security checks are made. All personnel shall abide by prevailing security requirements during visits during visits to such military property and shall obtain the required permits and passes in a timely manner. When applicable, the Contractor is responsible for preparation and submittal of application for all necessary passes and entry permits for Contractor and sub-Contractor personnel and vehicles to enter Department of Defense installations.

14. **Use of Microsoft Outlook meeting invitation requests**. The Contractor shall arrange for scheduled and ad-hoc teleconferences and meetings through use of Microsoft Outlook meeting invitations. As a standard, unless general guidance under this task order is provided by the TOCOR, Microsoft Outlook meeting invitations should contain the meeting location (or conference call-in number) and objective of the meeting, and should not include read ahead files directly attached to the invite. Read ahead files may be sent via email as reply-all to the meeting invite.

15. **Requests for information**.
    a. Requests for Government Furnished information shall be phrased in question form and be accompanied by the following information:
        i. A description of the specific information being requested.
        ii. Rationale why the Contractor is unable to obtain the information otherwise and why it is believed to reside within the Government.
        iii. Identify what document/analysis the information is needed for.
        iv. Why the information is needed, and when the information is requested to be provided by.
        v. How unavailability of the requested information will contribute to technical sufficiency (or lack of) for the effort it supports.

    b. The Government may request information from the Contractor in a similar fashion.

    c. Both Government and Contractor requests for information shall be recorded on the change/issue/request for information (RFI) log.

    d. As a standard, allow for at minimum one week for response to RFIs.

16. **Photography and use of photographs.** The Contractor is responsible to identify and comply with the appropriate procedures to obtain approval to take photos and for use of photographs in performance of work.

17. **GIS/CAD.** The Contractor is responsible to identify and comply with the appropriate procedures to obtain and use GIS/CAD electronic files in performance of work (see “Data Requirements” section for more information).

18. **Compliance with applicable requirements of the Navy's consultation with NMFS**. Work to be performed under this contract shall serve to meet the intent of and comply with the applicable portions of the Navy's consultation with NMFS regarding avoidance, minimization, and offset

mitigation of impacts associated with the proposed in-water, dredging, including standard procedures and best management practices applicable to work performed under this contract.

19. **OPSEC**. OPSEC requirements are applicable when Contractor personnel have access to or generate covered defense information (CDI) as defined in DFARS 225.204-7012. As such, OPSEC Measures (i.e., the planned action to conceal or protect identified critical information and indicators from disclosure, observation, or detection and to protect the same from collection) are applicable to this requirement. Contractor employees shall not discuss or disclose any information provided to them in the performance of their duties to parties other than authorized Government and/or Contractor personnel who have a "need to know" in accordance with the "Authorized Use and Non-Disclosure Agreement & OPSEC Certification of Understanding" executed by the Contractor which is incorporated in full by reference herein. Contractor shall not use any information or documentation provided by the Government or developed under this requirement for other purposes without the consent of the Government Contracting Officer IAW DFARS 252.04-7000. Contractor shall not release to anyone outside the Contractor organization any unclassified information, regardless of medium (e.g., film, tape, document, etc.), pertaining to any part of this contract or any program related to this contract, unless the Contracting Officer has given prior written approval IAW DFARS 252.204-7000. Markings: All deliverables/submittals generated by the Contractor shall be properly marked. For Official Use Only information generated and/or provided under this contract shall be marked in accordance with DoD 5200.01. Vol 4. Technical information shall also be marked with appropriate Distribution Statements and Export Control warnings in accordance with DoD 5230.24 and program Security Classification Guidance.

20. Certain information provided by the government may require unique handling, storage and or release/dissemination procedures. Contractors are cautioned to study "Authorized Use and Non-Disclosure Agreements & OPSEC Certification of Understanding" and comply accordingly.

21. **Note**: The following clauses have been incorporated in this solicitation/contract - DFARS 252.204.7000 Disclosure of Information, 252.204-7008 Compliance with Safeguarding Covered Defense Information Controls, and 252.204-7012 Safeguarding Covered Defense Information and Cyber Incident Reporting. DFARS Clause 252.204-7000 restricts the release of unclassified information outside contractor's organization without prior Contracting Officer permission, with exceptions; DFARS Clause 252.204-7008 requires Contractor compliance with Safeguarding Covered Defense Information Controls; and DFARS 252.204-7012 requires Contractor to provide adequate security for all covered Contractor information systems (including implementation of NIST 800-171 CUI requirements) and to comply with cyber incident reporting requirements.

VI. DATA REQUIREMENTS

a. File Format

Deliverables will be made by express mail and/or by electronic delivery. The Navy standard desktop computing software is Microsoft Office 2016. Final reports and other text documents shall be provided in Microsoft Word 2016 and Adobe Portable Document Format (PDF) readable with Adobe Acrobat Pro 2017 unless other mutually agreeable formats are determined. Spreadsheet files shall be provided in Microsoft Excel 2016 format. All graphics shall be delivered in Adobe Photoshop (.PSD), Adobe Illustrator (.ai), .JPG, .PNG, or original format that was used to develop the content. Files shall be delivered in a way that the content (including all layers and uncompiled/editable source files) can be used or updated for future efforts without needing to recreate the content or layers. All text, spreadsheet, graphics, database, and other files shall be delivered on CD-ROM, DVD, or other electronic media as approved by the technical COR (i.e., no flash drives). Database files shall be provided in Microsoft Access format, unless specified otherwise. All text, spreadsheet and database files shall be delivered on CD-ROM, DVD or other electronic media. Digital raster graphics will be submitted in high resolution >600 dpi Tagged Image Format (TIF); grid or cell-based digital raster spatial data will be submitted in ArcInfo Grid format or ASCII Grid format; spatial data shall be unprojected; the vertical reference elevation is Mean Sea Level (MLLW for bathymetry) with the relevant control data provided; and spatial data and metadata will be submitted together on CD ROM.

b. Geospatial Data, Maps, Drawings, and Sketches

I. Geographic Data Delivery Format

Digital geographic maps and the related vector data sets shall be developed using double precision and the data sets shall be delivered in a personal geo-database format that is Microsoft® SQL Server-compatible using ArcGIS 9.3.1 or higher and must be importable to a Microsoft® SQL Server geodatabase using ArcGIS Server 9.3.1 or higher. ArcInfo, ArcGIS, and ArcSDE are geographic information system software applications produced by the Environmental Systems Research Institute (ESRI) of Redlands, California.

II. Scale-dependent Mapping Guidelines

For installation and sub-regional level mapping, see NAVFAC Guidelines for Installation Mapping and GeoSpatial Data published by NAVFACENGCOM. Geospatial data for regional, theatre-level mapping (1:24K) shall be provided using precise geographic coordinates in decimal degree format with four decimal precision. Geographic data shall generally be provided in: nautical miles (nm.) for expansive marine areas, and statute miles (mi.) for expansive land areas, and projected into the Universal Transverse Mercator (UTM) coordinate system. Regional maps and data shall use the appropriate UTM Zone(s), the GRS 1980 spheroid and the North American Datum 1983 (WGS-84). No offsets shall be used. Each data set shall have a projection file if appropriate based on format. Map or drawing scales will be determined by the COR. Mapping accuracy for the agreed scales will conform to the American Society for Photogrammetry and Remote Sensing (ASPRS), "Accuracy Standards for Large-Scale Maps" and “Interim Accuracy Standards for Large-Scale Maps” (ASPRS, 1991). Copies of the ASPRS Accuracy Standards can be found online (http://www.asprs.org) or by contacting:

American Society for Photogrammetry and Remote Sensing
5410 Grosvenor Lane, Suite 210
Bethesda, MD 20814-2160.

III. Geographic Data Structure

The Contractor shall develop all geographic information in a structure consistent with the Spatial Data Standards for Facilities, Infrastructure, and Environment (SDSFIE), Version 2.6, released in January 2007, or a higher version if available at the time of this project. The Contractor shall consult with the Government concerning modifications or additions to the SDSFIE. The Government may approve modifications to the Standard if it is determined that SDSFIE does not adequately address subject datasets (which is the case in this task order). The American National Standards Institute's (ANSI) Committee for Information Technology Standards has approved the Spatial Data Standard for Facilities, Infrastructures, and Environment (SDSFIE) as ANSI standard NCITS 353. Copies of the SDSFIE can be found online (http://www.sdsfie.org/) or by contacting:

Director, U.S. Army Engineer Waterways Experiment Station
CADD/GIS Technology Center
Attn: CEWES-IM-DA/Smith
3909 Halls Ferry Road
Vicksburg, MS 39180-6199.

IV. Content Specifications for Geospatial Data

The Contractor shall develop all geographic information in a structure consistent with the latest version of Spatial Data Standards (SDS), unless otherwise specified by the COR. Specific content for environmental planning and range planning themes supplemental SDS is available from the NAVFAC Data Committee. The Contractor shall consult with the Government concerning modifications or additions to the SDS. The environmental data themes and attributes developed by the NAVFAC Data Committee have been specified to augment the existing standards in order to make geospatial data more useful to the specific business functions involved in environmental and ranges planning. Subject matter experts identified these themes to add value and relevancy to geospatial data deliverables.

V. Geographic Data Documentation

For each digital file delivered containing geographic information (regardless of format), the Contractor shall provide documentation consistent with the "Content Standards for Digital Geospatial Metadata, June 1998" published by the Federal Geographic Data Committee. The documentation shall include but is not limited to the following: the name and description of the map layer or coverage, the source of the data and any related data quality information such as accuracy and time period of content, description of equipment or instruments used in the data collection, the type of data coverage (point, line, polygon, etc.), the field names of all attribute data and a description of each field name, the definition of all codes used in the data fields, the ranges of numeric fields and the meaning of these numeric ranges, the creation date of the map layer and the name of the person who created it. A point of contact shall be provided to answer technical questions. Metadata generation tools included in the ArcGIS suite of software shall be used in the production of the required metadata in XML format. Alternately, another metadata generation tool called "document.aml" is available from ESRI for use with ARC/INFO to produce the required

metadata. If neither of these tools is used, the Contractor must insure that the metadata is delivered in a format, which can be easily translated to the XML format. All metadata needs to comply with the Federal Geographic Data Committee (FGDC) format. Copies of the FGDC metadata standard can be found online (http://www.fgdc.gov/metadata) or by contacting:

Federal Geographic Data Committee
590 National Center
Reston, Virginia 20192
USNG: 18STJ94731361, FAX: 703-648-5755, E-mail: fgdc@fgdc.gov.

c. Licensing and Ownership Rights

All technical and scientific data and computer software, including but not limited to source files, text files, graphics files, 2D and 3D imagery, new and existing video content, photographic content, audio content, animations, design documentation, other support documentation, GIS data, and other content developed by the Contractor are contemplated to be developed with Government funding and not Private funding. Therefore, unless the Contractor has made data rights assertions in accordance with the appropriate FAR/DFARS clauses, and the assertions have been accepted and made part of the contract by the Government, the Contractor shall provide all such technical data and computer software deliverables with Unlimited Rights.

VII. COMPLETION SCHEDULE AND DELIVERABLES

The Period of Performance for this contract task order is from the date of award through XXX. The Contractor shall adhere to the following schedule, unless otherwise approved by COR. Any deadlines not outlined in the table below shall be determined through discussion with the COR.

Item #	Task / Deliverable	Due Date*	Format
1	**Task 1: Kick-Off and Status Update Meetings**		
	Kick-Off Meeting	No Later Than (NLT) 2 weeks after contract award	Teleconference
	Project Schedule	NLT 2 weeks after kick-off meeting	Electronic
	1-hour Coordination Call(s)	As requested by Navy or Contractor	Teleconference
	Meeting Minutes	NLT 1 week after meeting is held	Electronic
	Monthly Status Update Calls	As requested by Navy or Contractor	Teleconference
2	**Task 2: Work Execution Plan, Dive Plan, and Health & Safety Plan**		
	Draft Work Execution Plan, Dive Plan, and Health & Safety Plan	NLT 4 weeks after kick-off meeting	Electronic
	Final Work Execution Plan, Dive Plan, and Health & Safety Plan	NLT 1 week after Navy review	
3	**Task 3: Biological Surveys**		
	Field Survey of Proposed Dredge Impact Areas	NLT than 8 weeks after the Navy approval of field plans	In-water field work
4	**Task 4: Deliverables**		
	Draft Survey Report	NLT 8 weeks following survey completion	Electronic
	Final Survey Report	NLT 3 weeks following Navy review	
	Orthomosaic Imagery	NLT 8 weeks following survey completion	External Hard Drive

*Durations are noted in calendar days (7 days per week regardless of weekend or holiday).